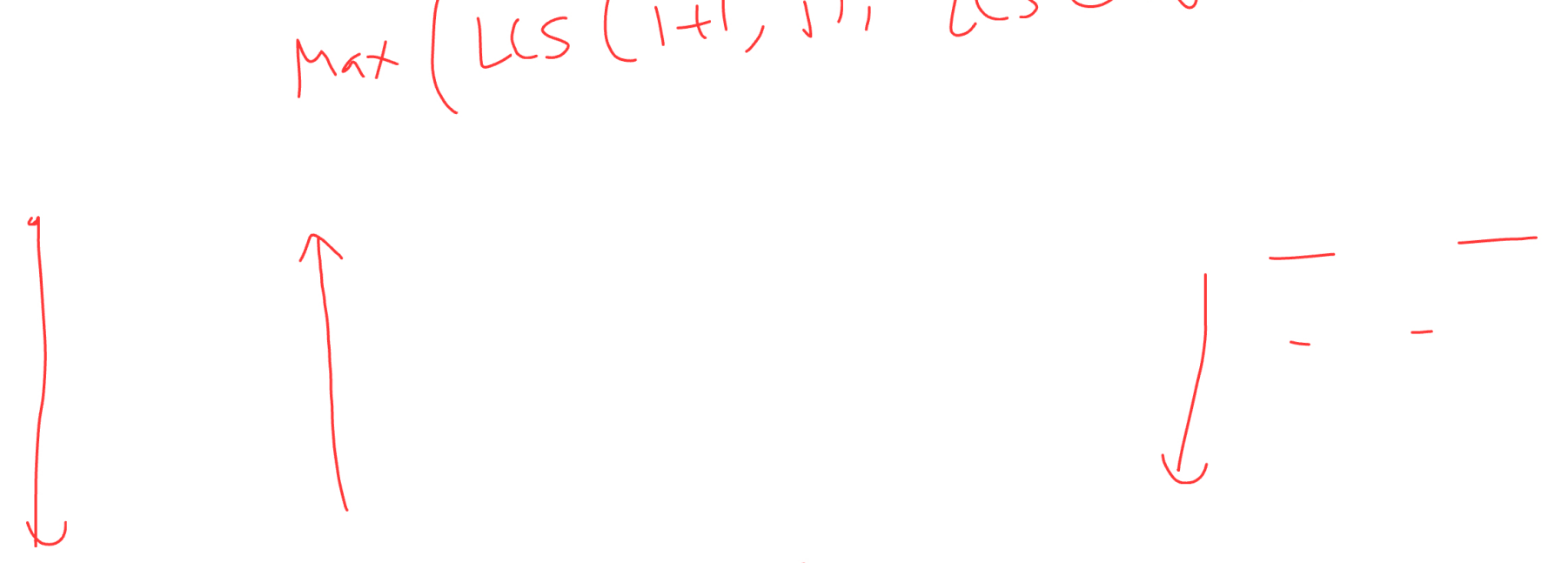
LCS

s1: ABCDGH
s2: AEDFHR
LCS = ADH

O/P = 3.

$LCS(i, j) = 1 + LCS(i+1, j+1)$ $\quad s1[i] == s2[j]$

$\max(LCS(i+1, j), LCS(i, j+1))$ $\quad$ otherwise.



Top-Down to Bottom-up.

1) Cater to the base-cases (logically).
2) Fill the rest of the table.

		0	1	2	3	4	5	6
		""	A	E	D	F	H	R
0	""	0	0	0	0	0	0	0
1	A	0	1	1	1	1	1	1
2	B	0	1	1	1	1	1	1
3	C	0	1	1	1	1	1	1
4	D	0	1	1	2	2	2	2
5	G	0	1	1	2	2	2	2
6	H	0	1	1	2	2	3	3

s2 (m+1)

s1 (n+1)

prev. Row.

BC

curr Row.

BC

(A, AE)

LCS (A, A)
= 1 + LCS("", "")

$dp[i][j] = LCS$ (First i chars of s1, First j chars of s2)

$= LCS(s1[0 \ldots i-1], s2[0 \ldots j-1])$

$dp[i][j] = 1 + dp[i-1][j-1]$ $\quad$ if $s1[i-1] == s2[j-1]$

$\max(dp[i-1][j], dp[i][j-1])$ $\quad$ otherwise

$(\overset{i}{A}BCDGH, \overset{j}{A}EDFHR) \rightarrow (i+1, j+1)$

$LCS(A, AE) = \max(LCS("", AE), LCS(A, A))$

Scope of optimization in AS:
while filling the i-th row, we only need the (i-1)th row. All the prev. rows (i-2, i-3 ... 0) are of no use.

[______ ___]

i → i-1, i-2 . i-3, i-4 ... i == 0
a b

0-1 Knapsack.

values: [60, 100, 120]
wts: [10/2, 20/2, 30/2]
W = 50

n.

→ 2^n combinations

I1 → [I1] [ ]

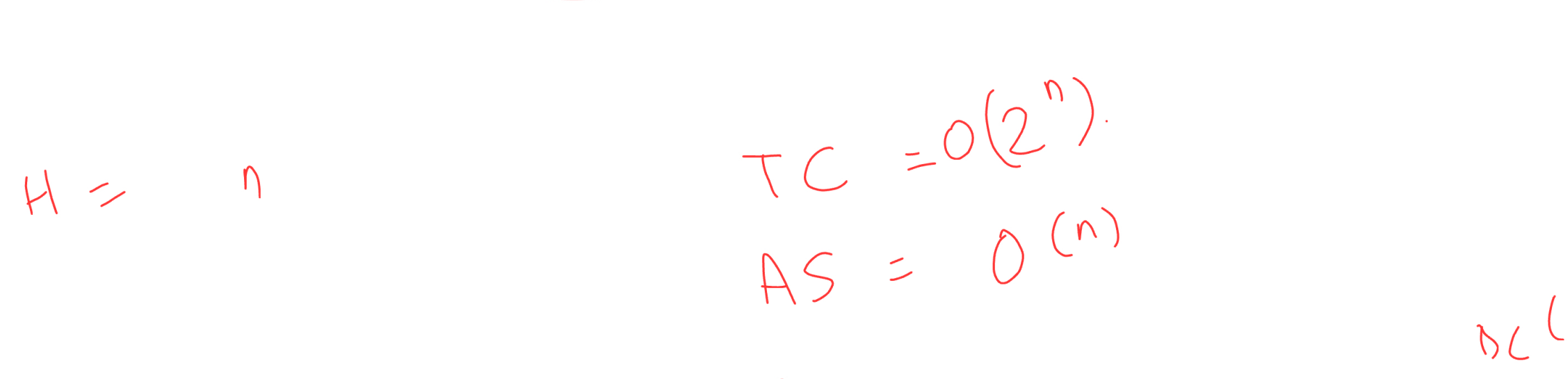
max(

$maxValue(i, W) = maxValue(i+1, W),$ $\quad$ Exclude the ith item

$val[i] + maxValue(i+1, W - wts[i])$ $\quad$ Include the ith item

)

$maxValue(i, W) = \max(maxValue(i+1, W), vals[i] + maxValue(i+1, W - wts[i]))$


n
(i, w)
min (n, W).
(i+1, W)
(i+1, W-1)
(i+2, w)
(i+2, W-1)
(i+2, W-1)
(i+2, W-2).

H = n

TC = O(2^n).
AS = O(n)

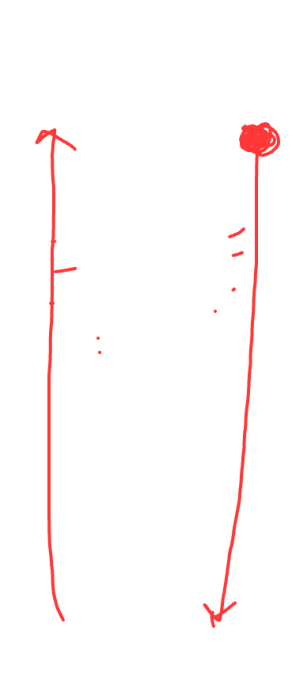
vals wts			0	1	2	....	10	...	50
		0 →	0	0	0		0		0
(60, 10)	I1	1	0	0	0		60	-----	60
(100, 20)	I2	2	0						
(120, 30)	I3	3	0						

BC (no items)

BC (W == 0)

$dp[i][j] = dp[i-1][j]$ $\quad$ Exclude.

$vals[i-1] + dp[i-1][j - wts[i-1]]$ $\quad$ Include.



Edit Dist.

A = abad
B = abac

abad → abacd → abac → 2
abad → aba → abac → 2
abad → abac → 1

A = Anshuman
B = Antihuman

Anshuman → Anthuman → Antihuman → 2

ED("", uman) = 4

ED(shuman, "") = 6

ED(Anshuman, Antihuman)
↓
ED(nshuman, ntihuman)
↓
ED(shuman, tihuman)

Insert: 1 + ED(shuman, ihuman)

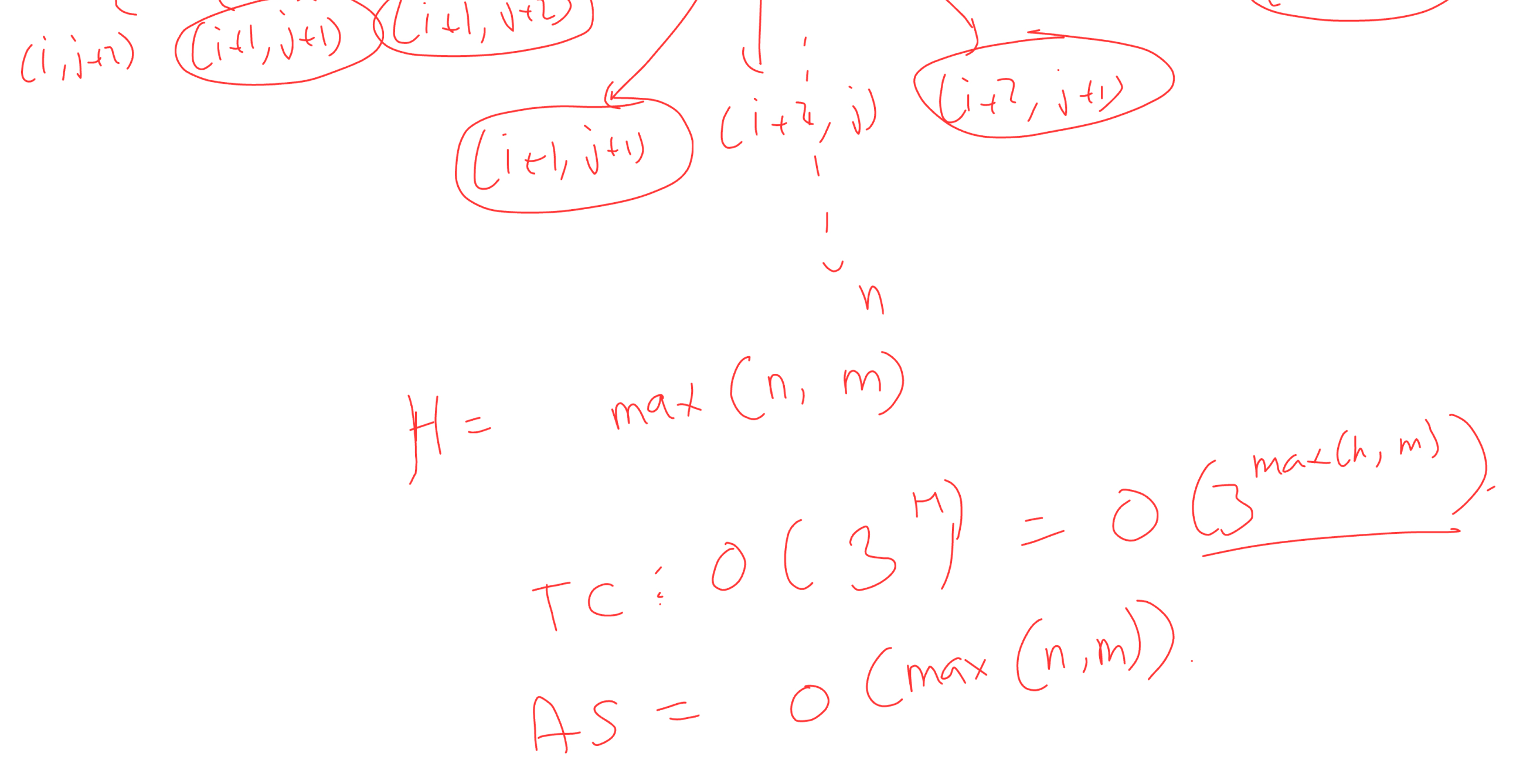
Delete: 1 + ED(human, tihuman)

Replace: 1 + ED(human, ihuman)

$ED(i, j) = ED(i+1, j+1)$ $\quad$ if $A[i] == B[j]$.

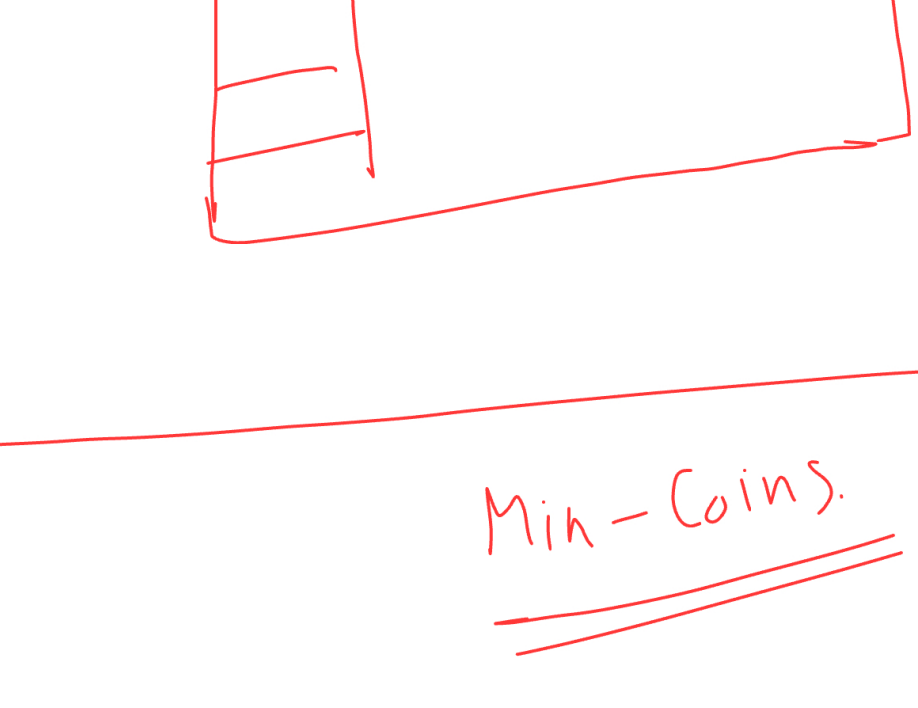
otherwise.

$1 + \min($
$ED(i, j+1)$ // insert.
$ED(i+1, j)$ // delete
$ED(i+1, j+1)$ // replace.
)


m
(i, j)
(i, j+1)
(i+1, j)
(i+1, j+1)
(i, j+2)
(i+1, j+1)
(i+1, j+2)
(i+1, j+1)
(i+2, j)
(i+2, j+1)
(i+1, j+2)
(i+2, j+1)
(i+2, j+2)
n

H = max (n, m)

TC: O(3^H) = O(3^{max(n, m)}).
AS = O(max (n, m))

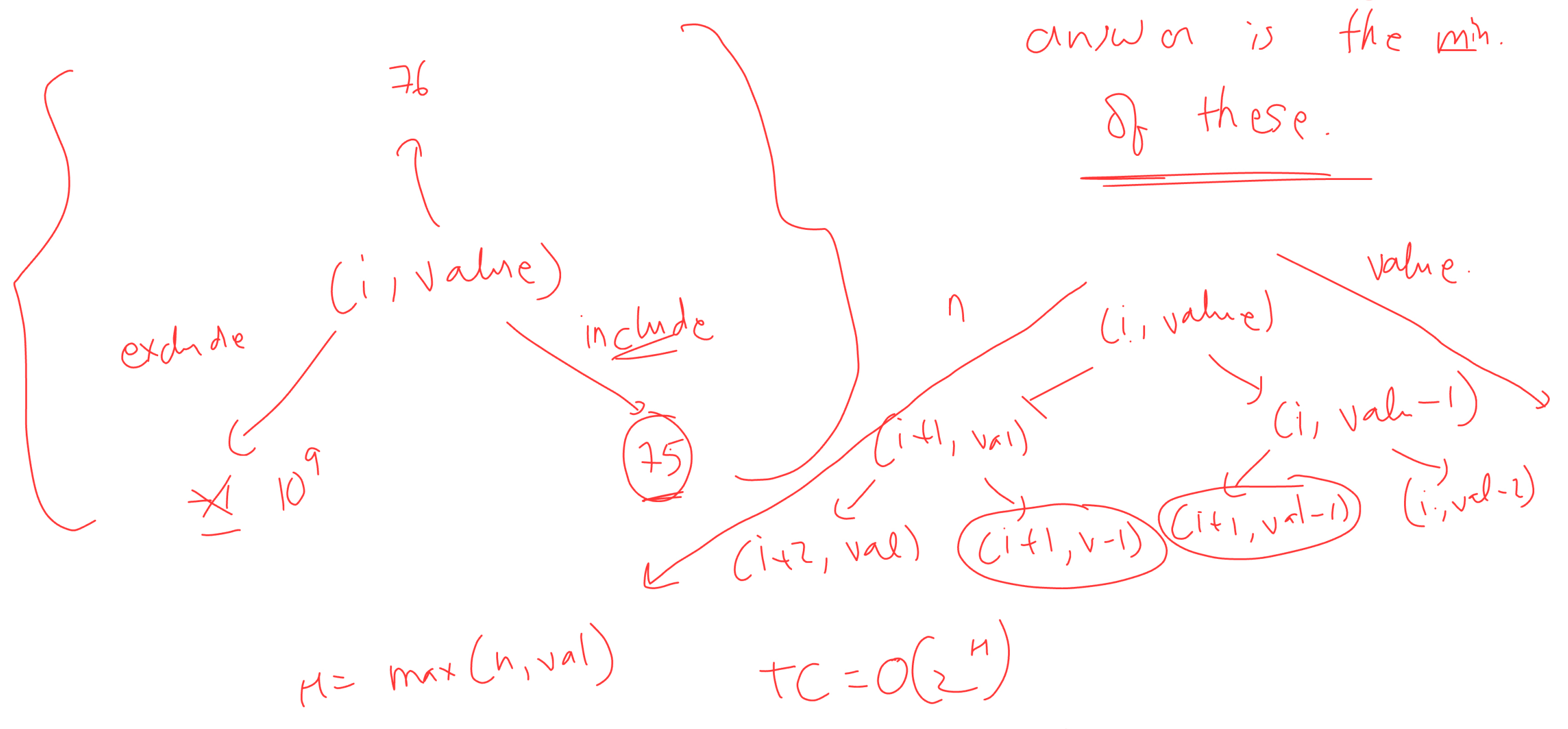


Min-Coins

$minCoins(value, i) = minCoins(value, i+1)$ $\quad$ Exclude

$1 + minCoins(value - coins[i], i)$ $\quad$ Include

answer is the min. of these.


76
(i, value)
exclude
include
10^9
75
n
value
(i, value)
(i+1, val)
(i, val-1)
(i+2, val)
(i+1, v-1)
(i+1, val-1)
(i, val-2)

H = max (n, val)

TC = O(2^H)
AS = O(H)